

Report:Arsh Corporation

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Contents

Contents	i
0.1 Project Title	1
0.2 Problem Statement	1
0.3 Dataset Collection	1
0.4 Methodology	1
0.5 Limitations	1

0.1 Project Title

AI-Powered OCR Extraction System

0.2 Problem Statement

Extracting important data manually from invoices and document images is time-consuming, inefficient, and susceptible to human mistakes. While conventional OCR systems can detect text, they frequently have difficulty in accurately extracting important details from various document formats and poor-quality images. This project seeks to create an AI-driven OCR system utilizing PaddleOCR and Google Gemini Pro to automatically identify text and extract vital information from images with precision and efficiency.

0.3 Dataset Collection

Various kinds of invoice and document images were gathered from publicly accessible datasets on Kaggle. The dataset contains diverse invoice formats, fonts, layouts, and image qualities to guarantee model resilience. These visuals were employed to train and evaluate the OCR system for precise text identification and information retrieval in various real-world situations.

0.4 Methodology

- To make user easily upload invoice or document images frontend is developed using HTML, CSS & Javascript.
- For better OCR performance the uploaded image is preprocessed using OpenCV by resizing and converting it into grayscale format.
- PaddleOCR is used to detect and recognize text from the processed image along with confidence scores.
- The extracted text is cleaned using regular expressions to improve text readability and formatting accuracy.
- Google Gemini Flash model analyzes the OCR text and extracts important information such as vendor name, invoice number, date, cash, and total amount in JSON format.
- Finally, the extracted information and confidence score are displayed on the web interface and can be downloaded as a CSV file using JavaScript.

0.5 Limitations

- Limited availability and access to Google Gemini Pro API due to quota restrictions.